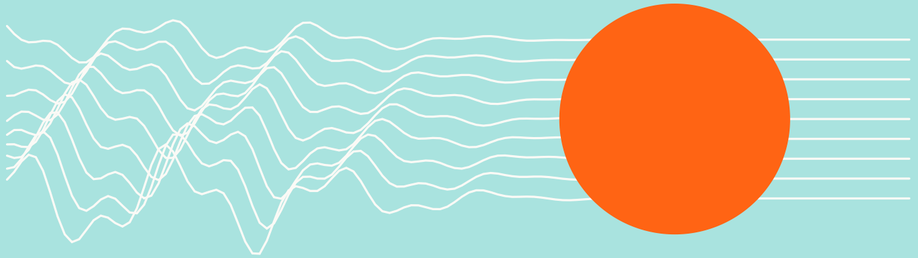
Reducing hallucinated transcripts in Whisper via hallucination space projection



Maryam Abbasihafshejani · Murtuza Jadliwala
The University of Texas at San Antonio
arXiv:2609.04561v1 [cs.AI] · 3 Sep 2026
Presented by: Nongmaithem Hans · Ravikant Sharma · AI553
Speech and Audio Processing, IIT Ropar · November 2026

Index

01 Introduction to the problem 3

02 The problem in depth 4

03 Proposed solution 6

04 The solution in depth 7

05 Results 12

06 Interpreting the results 14

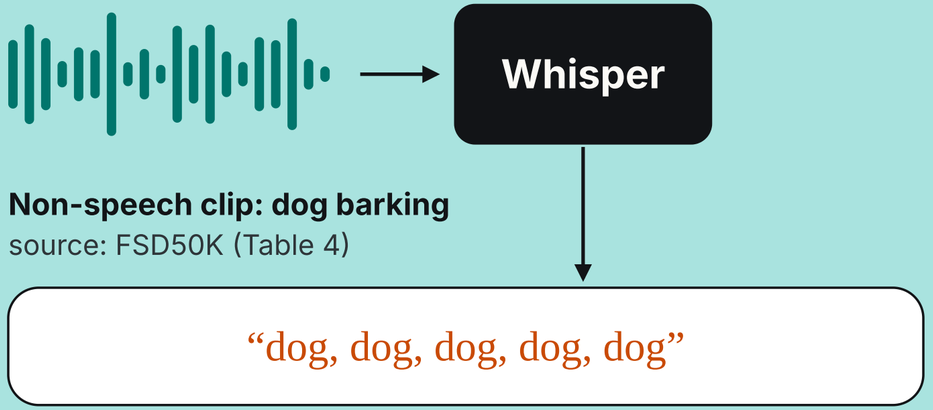
07 Our contributions 15

08 Results of our contributions 17

09 Summary and end 22

The problem: fluent text from no speech

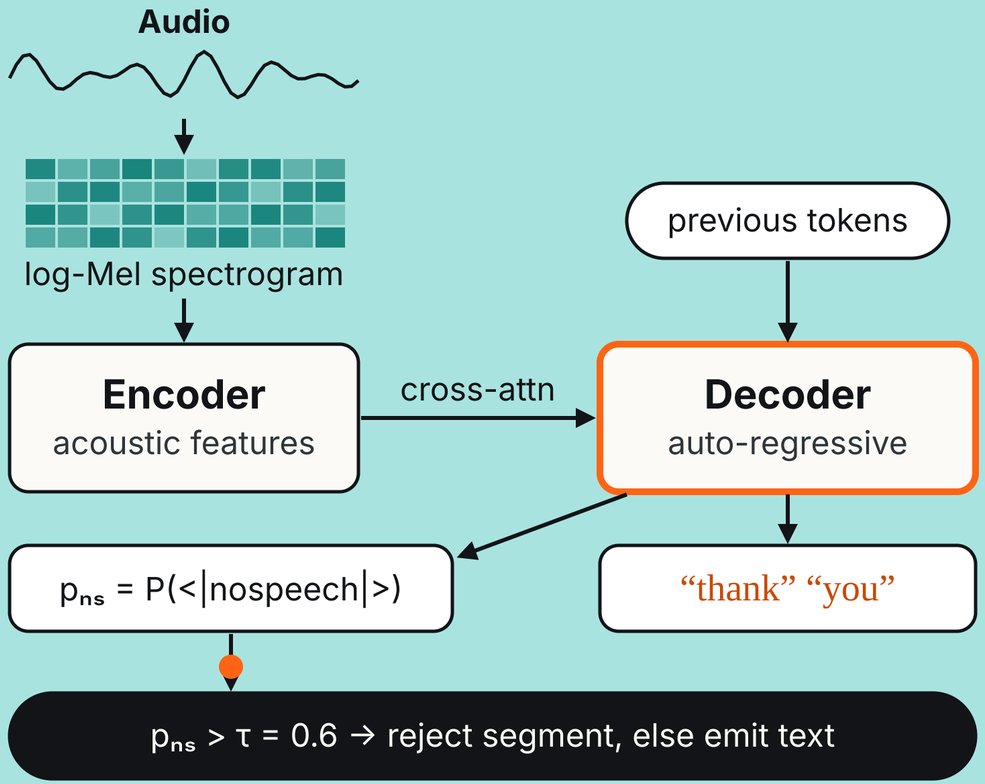
- Many audio segments (silence, music, environmental sounds, background noise) contain no speech, so the correct transcript is empty
- A reliable ASR system transcribes speech accurately and stays empty when there is none
- Whisper (Radford et al., 2023) is a state-of-the-art, widely used foundation model for ASR and robust speech processing
- Its generative sequence-to-sequence decoder can emit fluent text even when the input contains no speech
- For non-speech audio, any generated text is a hallucination
- Outputs look plausible (acknowledgments, subtitle-style endings, conversational fillers, repeated fragments), so downstream systems may treat them as real speech



Research question: can non-speech hallucinations be reduced by editing Whisper's decoder activations at inference time?

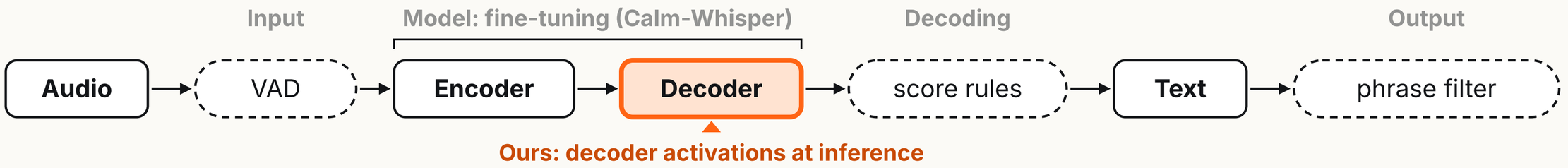
Background: how Whisper works

- Transformer encoder–decoder trained on 680K hours of weakly supervised multilingual, multitask audio; strong zero-shot performance across languages, domains and acoustic conditions
- Encoder maps audio to acoustic representations; the auto-regressive decoder generates text conditioned on the audio and previous tokens
- Transcription and no-speech estimation share the same decoder: there is no separately trained speech activity detector
- The decoder gives a probability to the special no-speech token <|nospeech|>, reported as no_speech_prob (p_{ns} in this deck)
- Rejection heuristic: if p_{ns} exceeds threshold τ (default 0.6), the segment is rejected as non-speech
- **Failure mode:** decoding is generative, so the model can still give high probability to fluent token sequences when acoustic evidence is weak or absent



Hallucination: fluent tokens even when there is no speech

Existing mitigations: where each one acts



Input · VAD

WhisperX (Bain et al., 2023) removes non-speech regions before decoding

Limit: needs an external segmentation model

Model · fine-tuning

Calm-Whisper (Wang et al., 2025) trains hallucination-related decoder parts on non-speech audio with empty transcripts

Limit: extra training; may overfit to its data

Decoding · heuristics

Thresholds on no-speech probability, log probability, compression ratio (Radford et al., 2023)

Limit: act only on output-level scores

Output · filtering

Bag of Hallucinations (Barański et al., 2025) removes known phrases after decoding

Limit: misses unseen hallucinations; can remove valid speech

Gap: none of these edit decoder activations at inference time. Hidden-state interventions already reduce hallucinations in LLMs and vision-language models (Yang et al., 2025; Dastmalchi et al., 2026).

Proposed solution: gated low-rank decoder projection

Training-free, at inference time: no parameter updates, no external VAD, no phrase lists. **Key idea:** hallucination-prone non-speech behaviour lives in a low-rank subspace of decoder hidden states that can be projected away.

01

Calibrate

Collect decoder hidden states for non-speech clips that hallucinate vs. clips correctly left empty

→

02

Estimate

SVD of their differences gives the top-r directions: the hallucination subspace, then kept fixed

→

03

Project

At one decoder layer, remove each hidden state's component along that subspace (strength α)

→

04

Gate

Apply it only when Whisper's own no-speech probability is at least γ

- **Two variants:** always-on (projection on every input) and gated (only when Whisper predicts likely non-speech)
- **Why the decoder:** it both generates text and provides the no-speech evidence, so the method acts before tokens are emitted

Step 1: calibration activations

Let ℓ be a decoder layer and d its hidden-state dimension. A non-speech calibration set gives two groups of layer-ℓ decoder hidden states:

$$H\ell = \{ h_{i,\ell}^{hall} \},\ i = 1 \dots N_h$$

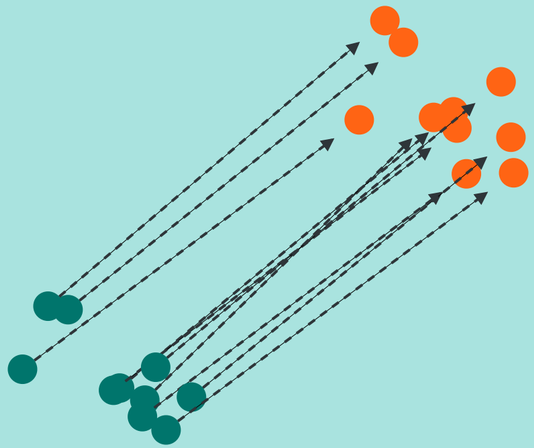
$$F\ell = \{ h_{j,\ell}^{empty} \},\ j = 1 \dots N_f$$

$$h_{i,\ell}^{hall},\ h_{j,\ell}^{empty} \in \mathbb{R}^d$$

- **Hℓ:** non-speech inputs where Whisper produces a non-empty, hallucinated transcript (N_h examples)
- **Fℓ:** non-speech inputs correctly rejected, with no transcript (N_f examples)
- Estimated only from the calibration set (ESC-50 folds 1–3), then kept fixed

Hℓ: non-speech, hallucinated (N_h)

Fℓ: non-speech, correctly empty (N_f)



$$\text{row } i \text{ of } \Delta\ell = (h_{i,\ell}^{hall} - h_{i,\ell}^{empty})^T$$

Illustrative 2-D sketch of layer-ℓ decoder states in $\mathbb{R}^d$

Step 2: low-rank subspace estimation

Both groups are non-speech, so differencing cancels shared acoustic factors and keeps the hallucination-related behaviour.

1 · Difference matrix, with n = min(N_h, N_f):

$$\Delta\ell \in \mathbb{R}^{n \times d},\ \text{row } i = \left(h_{i,\ell}^{\text{hall}} - h_{i,\ell}^{\text{empty}} \right)^{T}$$

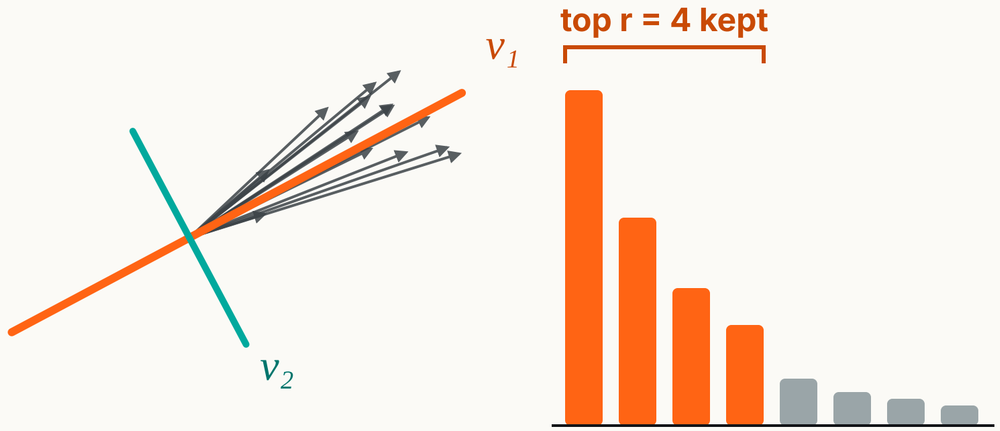
2 · SVD: the right singular vectors with the largest singular values are the dominant directions where the groups differ

$$\Delta\ell = U\ell\ \Sigma\ell\ V\ell^{T}$$

3 · Keep the top r as a row-orthonormal projection basis; rank r sets the dimension of the removed subspace

$$B\ell,r = V\ell,1{:}r^{T} \in \mathbb{R}^{r \times d}$$

$$\Delta\ell = U\ell\ \Sigma\ell\ V\ell^{T}$$



rows of Δℓ (difference vectors)

singular values $\sigma_1 \ldots \sigma_8$

$B\ell,r$ = top-r right singular vectors (orthonormal rows)

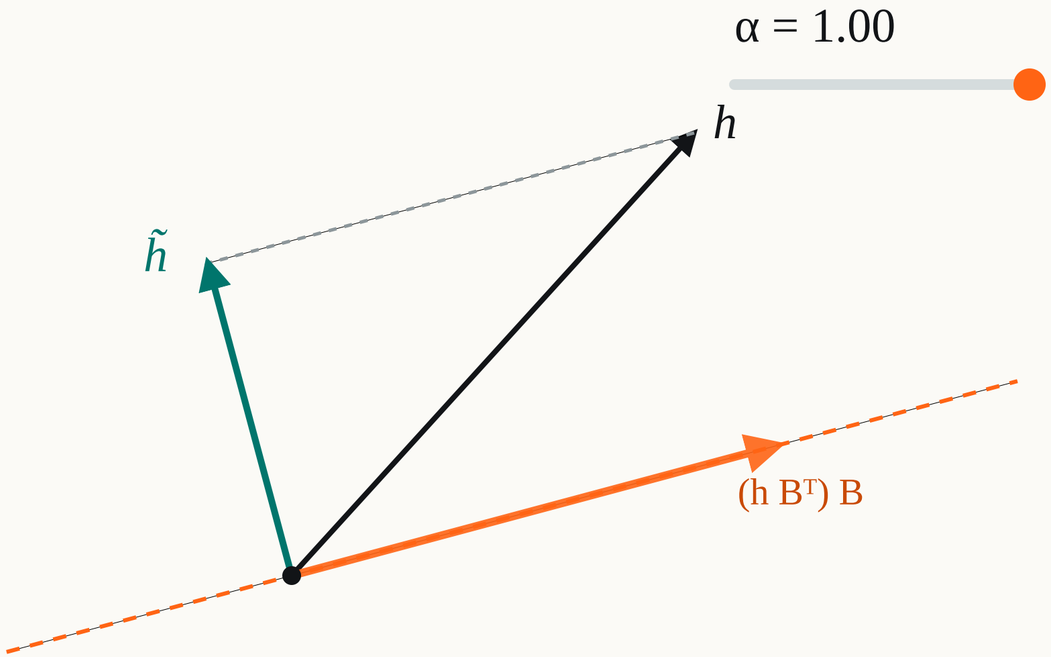
Step 3: inference-time decoder projection

A forward hook at the selected decoder layer replaces each hidden state $h_\ell \in \mathbb{R}^d$:

$$\tilde{h}_\ell = h_\ell - \alpha \left(h_\ell B_{\ell,r}^T \right) B_{\ell,r} \qquad (1)$$

$h_\ell B_{\ell,r}^T$	coordinates of h_ℓ along the r hallucination directions
$\left(h_\ell B_{\ell,r}^T \right) B_{\ell,r}$	the component of h_ℓ inside the hallucination subspace
α	projection strength; with orthonormal rows, $\alpha = 1$ removes the component fully, smaller α removes part of it

- **Intuition:** remove the directions that make non-speech inputs produce text, and keep the information normal transcription needs
- **No change** to model parameters or to the decoding procedure



Step 4: no-speech-conditioned gating

- Eq. (1) on every input can perturb real speech, so it is applied only when Whisper's own no-speech estimate is high
- p_{ns}^{base} = no-speech probability from a first, unprojected pass; γ = gate threshold

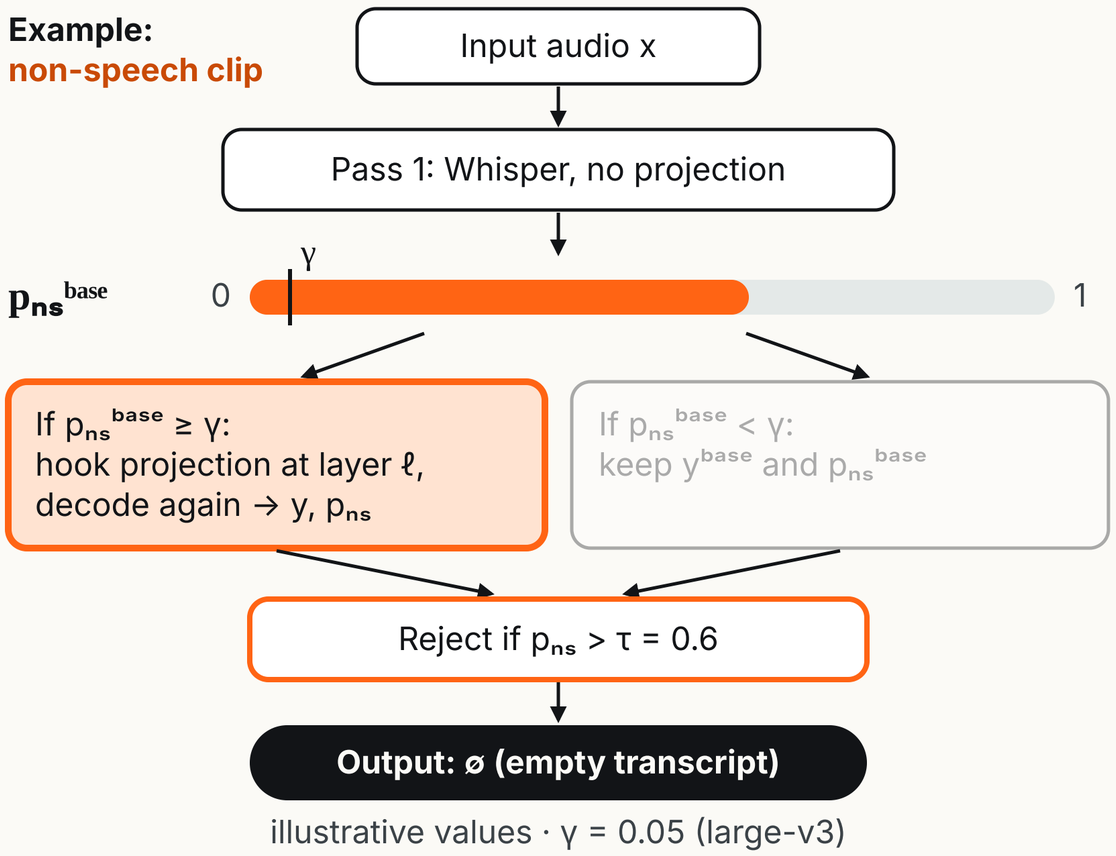
$$h'_\ell = h_\ell - \alpha \left(h_\ell B_{\ell,r}^T \right) B_{\ell,r} \quad \text{if } p_{ns}^{base} \geq \gamma$$

$$h'_\ell = h_\ell \quad \text{if } p_{ns}^{base} < \gamma$$

- Gate open: decode again with the projection; otherwise keep the unprojected output
- The resulting p_{ns} is then compared with τ for the final rejection decision

γ decides WHEN the activation intervention is applied

τ decides WHETHER the segment is rejected (default 0.6)



illustrative values · γ = 0.05 (large-v3)

Step 4: no-speech-conditioned gating

- Eq. (1) on every input can perturb real speech, so it is applied only when Whisper's own no-speech estimate is high
- p_{ns}^{base} = no-speech probability from a first, unprojected pass; γ = gate threshold

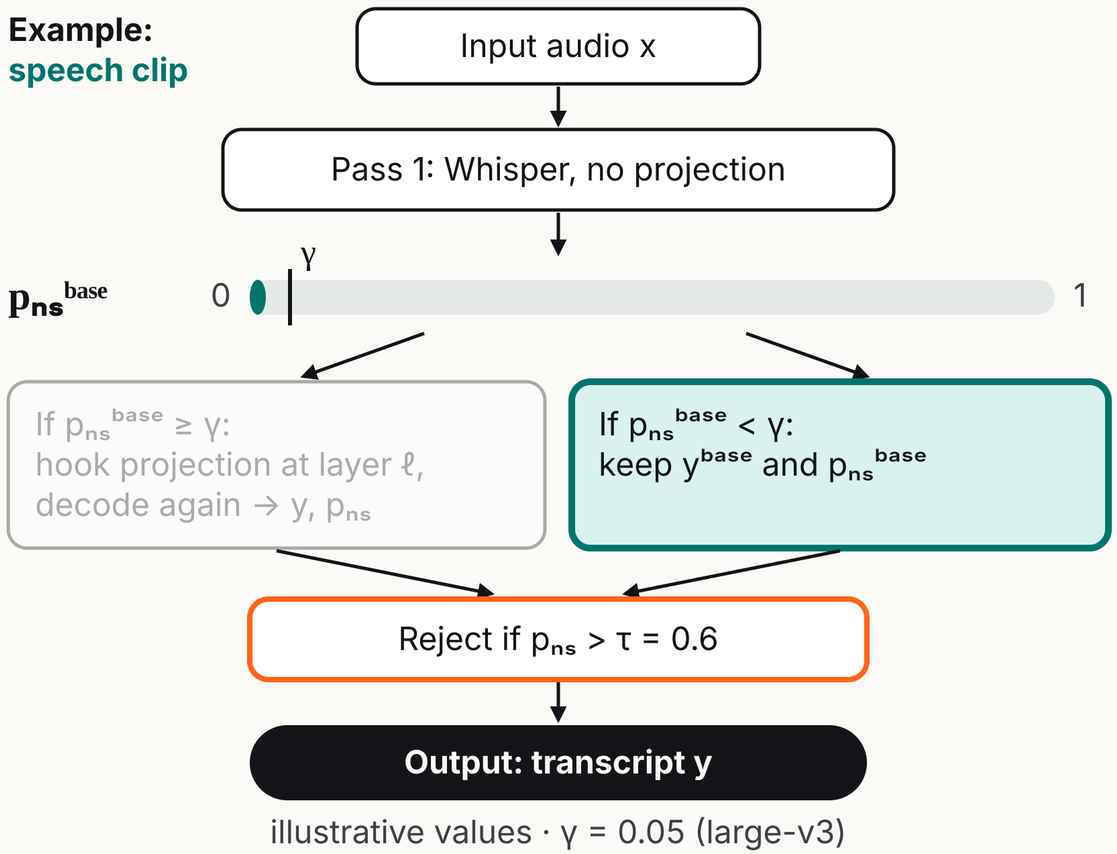
$$h'_\ell = h_\ell - \alpha \left(h_\ell B_{\ell,r}^T \right) B_{\ell,r} \quad \text{if } p_{ns}^{base} \geq \gamma$$

$$h'_\ell = h_\ell \quad \text{if } p_{ns}^{base} < \gamma$$

- Gate open: decode again with the projection; otherwise keep the unprojected output
- The resulting p_{ns} is then compared with τ for the final rejection decision

γ decides WHEN the activation intervention is applied

τ decides WHETHER the segment is rejected (default 0.6)



Evaluation: metrics and parameters

HR

Hallucination rate: non-speech inputs whose output is still non-empty after no-speech filtering (reference is empty)

WER

Word error rate on LibriSpeech, the cost to real transcription

FRR

False rejection rate: fraction of speech samples wrongly rejected by the no-speech filter

Four parameters, one forward hook on one decoder layer

ℓ	r	α	γ
decoder layer	rank of the subspace	projection strength	gate threshold

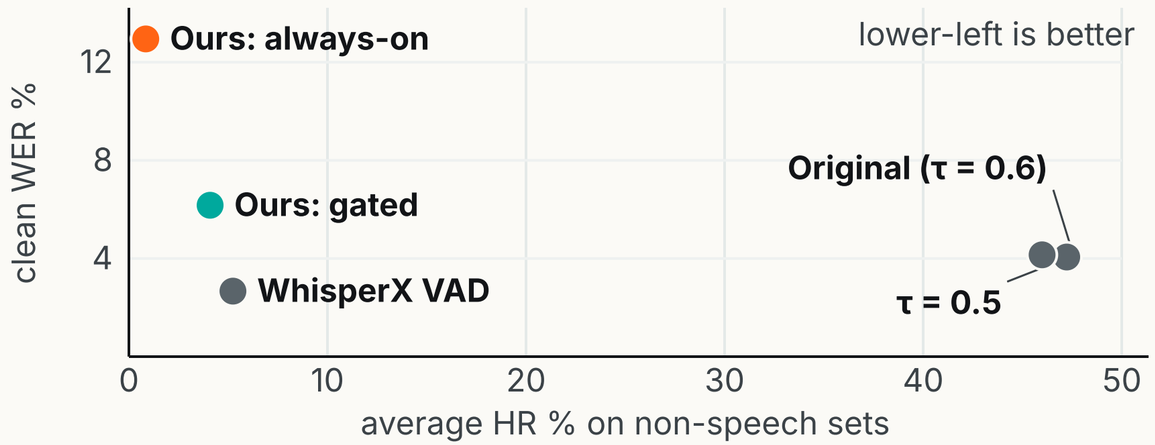
Whisper's default no-speech threshold $\tau = 0.6$ throughout. Parameters are chosen per Whisper scale (different depth and hidden size), once, on ESC-50 folds 1–3 plus LibriSpeech validation-clean (only to control speech degradation), then **fixed** for every held-out evaluation.

Result 1: non-speech hallucination rate (HR, %)

Model	Method	ESC-50	UrbanSound8K	FSD50K
Small	Original	23.50	12.33	21.35
	Always-on	1.50	0.81	6.68
	Gated	1.53	0.84	6.84
Medium	Original	26.50	14.52	41.95
	Always-on	1.82	0.62	8.02
	Gated	2.75	0.66	8.78
Large-v3	Original	44.25	76.08	21.35
	Always-on	1.50	0.87	0.18
	Gated	8.38	2.74	1.15

ESC-50 = held-out folds 4–5 · Whisper no-speech threshold $\tau = 0.6$ · lower is better

Versus baselines: the full trade-off



- Threshold tuning alone barely moves HR (47.23% → 45.99%)
- WhisperX keeps the best speech quality, but needs an external VAD and segmentation stage
- Gated lowers average HR further with no external detector, at a modest WER and FRR cost; always-on gives the lowest HR at a larger cost

Method (large-v3)	Avg HR	Clean WER	Other WER	Clean FRR	Other FRR
Original (τ = 0.6)	47.23	4.06	5.87	0.04	0.00
Threshold tuning (τ = 0.5)	45.99	4.15	5.87	0.08	0.00
WhisperX VAD	5.24	2.67	4.74	0.00	0.00
Gated projection	4.09	6.17	6.57	2.86	1.40
Always-on projection	0.85	12.95	13.13	10.50	11.47

What it means: the hallucination results

01 **Hallucination is low-rank.** A fixed subspace of 1 to 4 directions at one decoder layer pushes non-speech inputs over the rejection threshold.

---

02 **It transfers across datasets.** Calibrated on ESC-50 folds 1–3 and tested on folds 4–5, UrbanSound8K and FSD50K, which suggests a reusable decoder-level signature, not ESC-50-specific acoustics.

---

03 **Depth matters most, rank saturates.** Early layers are weak, middle-to-late layers strong; more directions are not better and can remove speech-relevant information.

---

04 **The baseline problem is severe.** Original large-v3 hallucinates on 76.08% of UrbanSound8K clips: 0.87% always-on, 2.74% gated.

---

05 **Always-on vs. gated.** Always-on gives the lowest HR; gated is slightly higher on large-v3 (8.38% vs 1.50% on ESC-50) but protects speech. For small they are nearly identical (1.53% vs 1.50%).

Our contributions: reproduce, then test it

01 **Calibrate**

1,200

clips, ESC-50 folds 1–3
256 hallucinated · 944 empty

→

02 **Fit bases**

12 × 4

layers × ranks (1, 2, 4, 8)
SVD of paired differences

→

03 **Select**

48 + 16

configs: ℓ × r, then α × γ
dev WER ≤ plain + 0.5 pts

→

04 **Evaluate**

6,800

test clips, 5 sets, 7 modes
paired McNemar tests

Setup

- Whisper small, fp16, on an RTX 3050 6 GB laptop GPU; about 3 hours end to end
- Greedy decoding, English forced, 30 s window; reject when $p_{ns} > \tau = 0.6$
- Seeded test subsets: 800 ESC-50, 2,000 UrbanSound8K, 2,000 FSD50K, 1,000 + 1,000 LibriSpeech

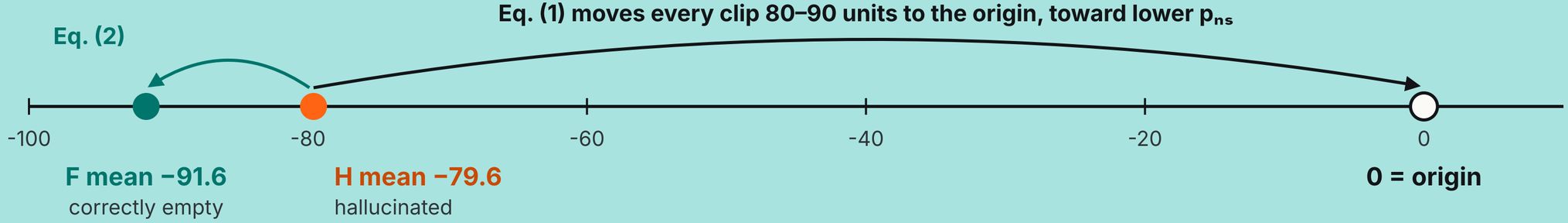
Four contributions

1. **Reproduction** on Whisper small, including the paper's own operating point
2. **Threshold baseline** the paper omits: τ chosen on dev by the paper's own rule, compared clip by clip
3. **Analysis** of what the gate and the edit do, and why the last layer fails
4. **Affine variant:** move states to the empty-class mean instead of zero

Why the last layer fails: and our affine fix

Coordinate of the layer-12 SOT state on the first basis vector (calibration clips)

it correlates with log p_{ns} at −0.73: the direction is largely Whisper's own no-speech read-out



Paper, Eq. (1): project to zero

$$\tilde{h} = h - \alpha\,(h\,B^{T})\,B$$

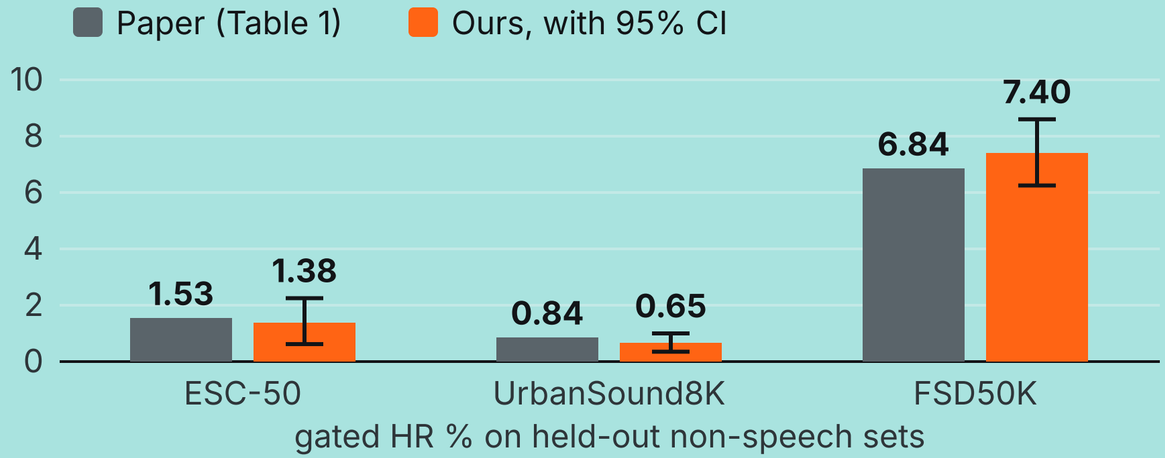
At layer 12, clips with $p_{ns} > 0.6$ drop from **78.7% to 0%** at every rank: nothing is rejected, so HR is 100%

Ours, Eq. (2): move to the empty-class mean

$$\tilde{h} = h - \alpha\,((h - c)\,B^{T})\,B$$

c = mean of the F states at layer ℓ. Clips with $p_{ns} > 0.6$ rise from 78.7% to **95.2%** (r = 1) and **99.4%** (r = 8)

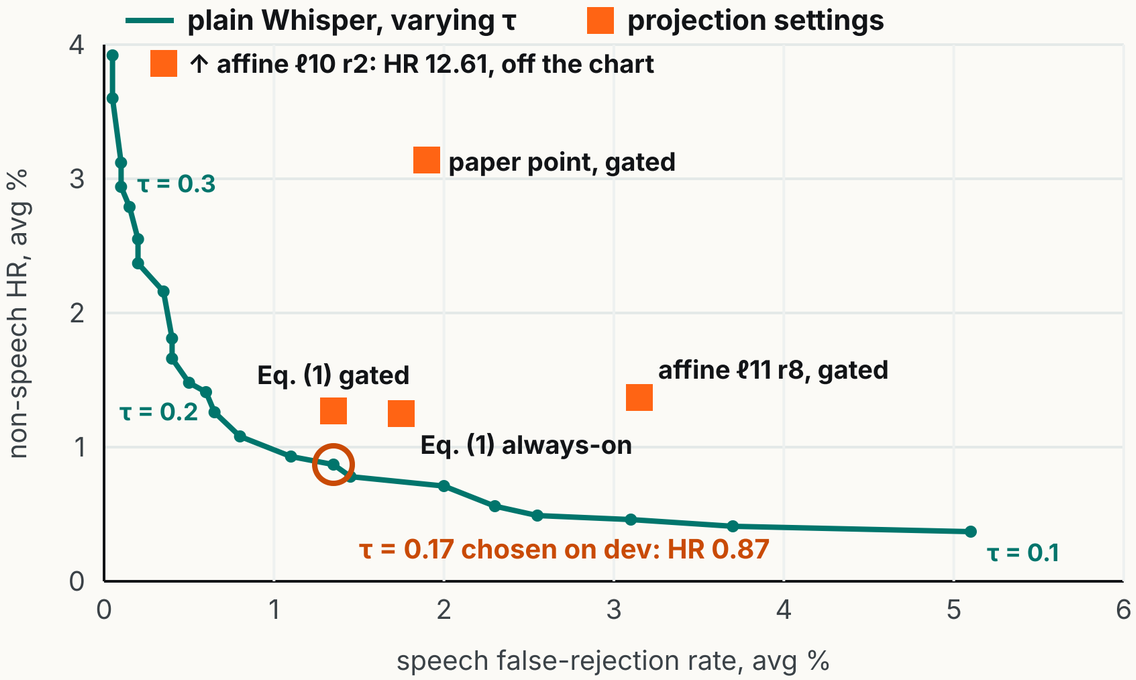
Reproduction: the paper's own setting on Whisper small



- **HR reproduces:** gated mean 3.14% against the published 3.07%; all six values inside our 95% CIs
- **Gated speech cost is close:** WER +0.60 / +0.78 points over plain Whisper, against the paper's +0.33 / +0.75
- **Always-on does not:** FRR 6.7% / 14.7% against 0.64% / 4.86%, so 4–10× more speech is lost

ℓ = 10, r = 1, α = 1, γ = 0.15	HR ESC-50	HR US8K	HR FSD50K	WER clean	WER other	FRR clean	FRR other
Original · paper	23.50	12.33	21.35	4.04	8.38	0.00	0.00
Original · ours	26.12	14.50	**44.60**	3.31	7.63	0.00	0.00
Always-on · paper	1.50	0.81	6.68	4.51	11.48	0.64	4.86
Always-on · ours	1.38	0.55	7.10	**8.20**	**19.49**	**6.70**	**14.70**
Gated · paper	1.53	0.84	6.84	4.37	9.13	0.41	2.58
Gated · ours	1.38	0.65	7.40	3.91	8.41	**1.40**	2.40

A tuned threshold: matches or beats the projection



Held-out, Whisper small	HR	WER	FRR
Plain, τ = 0.6	28.41	5.47	0.00
Plain, τ = 0.17 (dev)	**0.87**	**5.93**	1.35
Gated, Eq. (1) ℓ11 r8	1.27	6.58	1.35
Gated, paper point	3.14	6.16	1.90
Gated, affine ℓ10 r2	12.61	5.68	0.35

- Same selection rule, a τ grid of step 0.01: on dev, τ = 0.17 gives HR 0% at unchanged WER
- No projection lies below or left of the threshold curve
- p_{ns} already separates speech from non-speech: AUROC 0.9991 plain, 0.9986 gated
- The paper's τ sweep stops at 0.5

At equal HR: projection rejects more speech

Paired test vs plain Whisper at the same HR	Speech rejected (proj. / τ)	McNemar p	Hallucinated (proj. / τ)	p	Avg WER diff [95% CI]
Gated Eq. (1) ℓ11 r8 vs τ = 0.2	**24** / 10	**0.024**	45 / 46	1.0	+0.95 [0.40, 1.56]
Gated, paper point vs τ = 0.3	**36** / 0	**$3 \cdot 10^{-11}$**	91 / 91	1.0	+0.68 [0.40, 0.99]
Gated affine ℓ10 r2 vs τ = 0.46	**7** / 0	**0.016**	192 / 169	0.25	+0.21 [0.06, 0.41]

What the gate does at the paper's point (γ = 0.15)

	utterances		gate open		then rejected
test-clean	1,000	→	14	→	14
test-other	1,000	→	26	→	24

On speech the method is the threshold "reject if $p_{ns} \geq 0.15$"

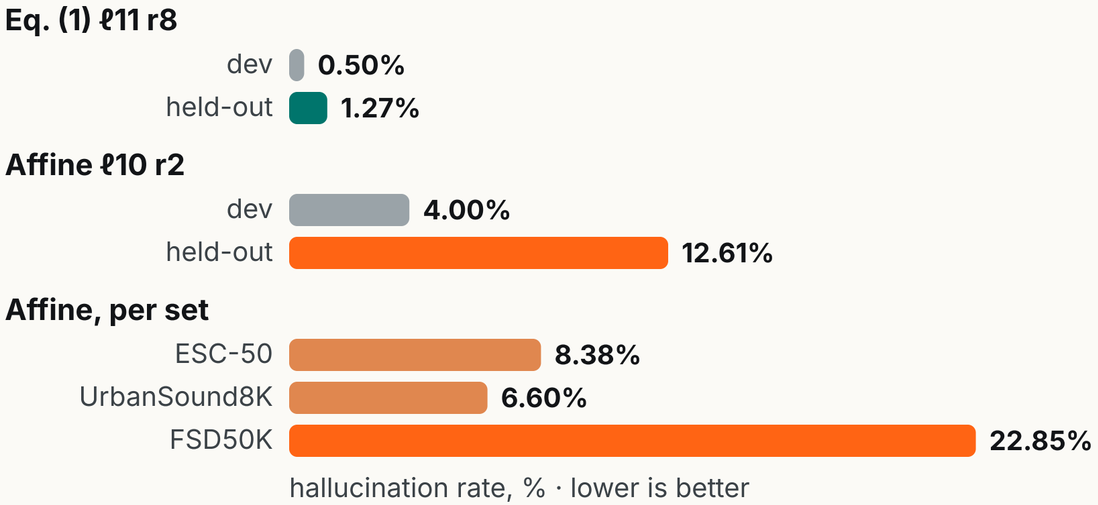
Same as plain τ = 0.15

WER 3.91 / 8.45, FRR 1.4 / 2.6, against 3.91 / 8.41 and 1.4 / 2.4, with lower HR (0.71% vs 3.14%)

The cost is rejection only

On projected, accepted utterances WER barely moves (4.55 → 4.65 clean); rejected ones are short (9.1 vs 20.2 words)

Affine variant: keeps speech, misses new sounds



- **Keeps speech:** FRR 0.35% and WER 5.68%, against 1.35% and 6.58% for Eq. (1)
- **Generalises poorly:** held-out HR 12.61% against 4.0% on dev; FSD50K sound classes are not in ESC-50
- **Still loses to a threshold:** at equal HR ($\tau = 0.46$) it rejects 7 utterances, the threshold 0 ($p = 0.016$)

Why it breaks at other settings

c is the mean at the SOT position, but the hook edits every position. At ℓ11 r8 the decoder emits only "<|zh|><|zh|>…": WER 99.6% on edited utterances

A side effect we do not count

Always-on affine lowers test-clean WER 3.31 → 2.92 (−0.39 [−0.77, −0.09]) but not test-other (+0.24 [−0.17, 0.69])

What this means: issues in the paper's evaluation

1. **The key control is missing.** The τ sweep stops at 0.5, though the paper's own medium ablation shows plain HR 29.49% → 13.08% from τ 0.6 to 0.4

2. **At the published small setting the edit adds nothing on speech:** every gated utterance is rejected

3. **WER mixes two errors.** On accepted utterances the edit does no harm; the whole cost is rejecting short utterances

4. **Projecting to zero is not neutral.** An affine target fixes the last layer, but one SOT centre at every position breaks decoding

5. **Labels come from p_{ns},** so the subspace partly re-learns the no-speech read-out, consistent with unchanged AUROC

6. **Tuning ignores noisy speech:** our Eq. (1) setting had 0% FRR on dev-clean but 2.5–3.3% on test-other

7. **Internal inconsistencies:** validation WER 3.94% vs 11.59%, medium α 0.75 vs 1.0, FSD50K 21.35% for two models, the FSD50K count

Limits of our study: Whisper small only, one seed, held-out subsets, no WhisperX VAD run

In summary: seven things to remember

Problem	Whisper hallucinates fluent text on non-speech audio (large-v3: non-empty on 44.25% of ESC-50 and 76.08% of UrbanSound8K clips)
Method	Training-free, low-rank decoder activation projection, calibrated once on non-speech data and fixed, optionally gated on Whisper's own no-speech probability
Results	Average HR 31.31% → 2.44% (always-on) and 3.74% (gated): relative reductions of 92.21% and 88.05%
Cost	Gated WER +0.33 to +4.39 points and FRR 0.41% to 9.97%; the HR–WER–FRR trade-off is controllable through τ, γ and α
Vs. baselines	Large-v3 gated average HR 4.09% beats WhisperX VAD (5.24%), BoH (44.09%) and threshold tuning (45.99%), with no external detector, phrase list or fine-tuning
Our reproduction	On Whisper small the paper's HR reproduces (3.14% vs 3.07%), but plain Whisper with $\tau = 0.17$ chosen on dev does at least as well (HR 0.87%, WER 5.93%, same FRR); at equal HR every projection rejects more speech
Our affine variant	Moving states to the empty-class mean fixes the last-layer failure and nearly removes false rejections, but it generalises poorly (held-out HR 12.61%)

Thank you

Questions?

Reducing Hallucinated Transcripts in Whisper via Hallucination Space Projection · arXiv:2609.04561v1

